

# **DIT5307 Brief Development Report: THEi HR System**

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# 1. Introduction

The **THEi HR System** is a **Java EE 8** enterprise web application developed to streamline internal human resource administration. The system provides an administrative portal that supports: (1) employee profile management, (2) training program enrollment management, (3) installment-based fee payment recording, and (4) automated compliance monitoring through scheduled fee-warning generation.

## 2. Business Logic Layer (Enterprise Session Beans)

The business logic layer is implemented using **EJB** to ensure consistent lifecycle handling, scalability, and reliable container services (transactions, scheduling, and pooling).

### 2.1 **CrudServiceBean** — Stateless Session Bean

**CrudServiceBean** functions as the primary service layer for handling database operations. It performs localized **CRUD**

### 3. Transaction Management

This project relies on **Container-Managed Transactions (CMT)**, which are provided by the EJB container and applied to business methods that modify persistent data.

Transactional boundaries are used for all write operations in `CrudServiceBean` (e.g., creating employee records, updating enrollments, inserting payment records, and generating warnings). This ensures that related changes across multiple entities are committed atomically and remain consistent.

## 4. Presentation Layer (JSF Pages and Backing Beans)

The user interface is built using **JSF (Facelets)** and styled with **Bootstrap 5** for a clean layout and improved usability. JSF backing controllers are implemented primarily using `@ViewScoped` and `@RequestScoped` scopes to balance responsiveness and state management.

### 4.1 Authentication

- `login.xhtml` + `LoginController`

Serves as the system entry point. It

## 5. System Design (Persistence Model)

The domain model is implemented using JPA entity relationships reflecting core HR and training-payment requirements:

- Employees can participate in multiple training programs.
- Training programs can include multiple employees (via enrollments).
- Each enrollment maintains payment-related state (total fee, paid-to-date, outstanding balance).

## 6. Development Assumptions

To bridge unspecified requirements in the brief, the following assumptions were adopted:

1. **Authentication Scope:** A single administrative account ( `hr` ) is sufficient for demonstration, rather than implementing full role-based access control (RBAC).
2. **Demonstration-Friendly Scheduling:** Timer intervals were shortened (e.g., seconds instead of real-world days) to

## 7. References

1. Apache Software Foundation, "Apache Maven project documentation," 2023.  
Available: <https://maven.apache.org/>
2. Oracle, "Java EE 8 Technologies (EJB 3.2, JPA 2.2, JSF 2.3, etc.)." Available:  
<https://www.oracle.com/java/technologies/java-ee-8.html>
3. Oracle, "Java EE 8 SDK Downloads (includes GlassFish 5.0.1)." Available:  
<https://www.oracle.com/java/technologies>